

TUTORIAL TWO

Processes and Threads

1. Indicate whether the following statements are true or false. Justify your answers.
 - a) A ready process waiting to get access to the CPU is in the “waiting” state.
 - b) When any hardware interrupt (not a trap) arrives, the process that is currently in the “running” state (if any), will be moved to the “ready” state.
 - c) The “wait()” system call is generally used by a child process to wait for instructions from a parent process.
 - d) Message passing based Inter-Process Communication (IPC) consumes less memory than shared memory based IPC.
2. What are two main differences between the data and stack regions of a process memory?
3. Explain the difference between a single-threaded and a multi-threaded process.
4. The figure below shows the execution of processes P0 and P1 in a multiprogramming system.
 - a) Identify state transitions of each process.
 - b) Describe operations A, B, C and D performed by the operating system kernel.

